Lecture 0
CSC1104 Course Overview

CSC1104: Data Warehousing and Data Mining
2026/2027 Semester 1

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Course Structure
Continuous Assessment
Exam & Resources

Agenda

1. Course Structure
2. Continuous Assessment
3. Exam & Resources



Course Structure
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Exam & Resources

CSC1104 Module Structure

- Exam Component: 75%
- 5 sections covering data mining (ML, AI) and data warehousing (engineering)
- Coursework Component: 25%
- Data Mining Project
- 2 parts: workshops & final report



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Section 1: Classification

- Bayes Classifiers
- Nearest Neighbour
- Decision Trees
- Attribute Selection Measures



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Section 2: Model Performance

- Evaluation Measures
- Comparing Classifiers
- Ensemble Models

Section 3: Cluster Analysis

- k-Means Clustering
- k-Medoids Clustering
- Hierarchical Clustering

Section 4: Association Rule Mining

- Association Rules
- The Apriori Algorithm
- Generating Itemsets
- Generating Rules

Section 5: Warehousing and OLAP

- Data Cubes and OLAP
- Warehouse Design and Usage
- Warehouse Architecture
- Warehouse Implementation

Section 6: Data Cube Computation

- Efficient Computation of Data Cubes
- Indexing OLAP Data
- Efficient Processing of OLAP Queries
- Materialising Different Cube Types
- Strategies for Cube Computation

Project Goal: Make a prediction

- You need a trained dataset
- You need an idea (prediction)
- You use 80% of your data to *learn*; you test with the rest
- 2-person teams
- Some ideas . . .

Workshop 1: Pitch Your Idea

- Week 4: 60-second pitch for your project (all teams)
- 1 or 2 slides
- Describe your dataset and what you are hoping to do
- No Marks . . . just a feedback session
- Some students reported this as the most important session!

Workshop 2: Learn about your Data

- Counts for 30% (of 25%).
- Option 1: Measuring the Central Tendency
- Option 2: Measuring the Dispersion of Data
- Option 3: Graphic Displays
- Option 4: Data Cleaning
- Option 5: Integration or Transformation
- Implementation will depend on the final students numbers . . .

Project

- Remaining 70% (of 25%).
- Deliverable is a Report & Dataset
- Generate your own dataset (using a web scrape) or use an online dataset
- Classification (prediction) problem



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Report & Marking Structure

- S1: Idea and dataset description . . . 20%
- S2: Data Preparation (from the independent learning) . . . 25%
- S3: Algorithm description . . . 5-15%
- S4: Results and Your Analysis (insights!) about what you found . . . 40-50%



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Exam Structure

- 3 hours
- Answer 4 from 5



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Resources

- Data Mining: Concepts and Techniques (Han & Kamber), Morgan Kaufmann (3rd edition)
- Chapter 2 covers the data analysis workshop (PDF on Loop)
- Exam is covered in chapters 3,4,5,6,7,8,12.



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Lecture & Assessment Schedule



Week	Workshops (Tue)	Lectures (Wed)	Notes	Attendance
Week 1	No workshop	Introduction & Coursework	Coursework starts in Week 1!	Wednesday only
Week 2	No workshop	Classification i		Wednesday only
Week 3	No workshop	Classification ii		Wednesday only
Week 4	*Pitch Your Project*	Model Performance	**All Project Groups Present**	Tues & Wed
Week 5	*Data Analysis*	Cluster Analysis i	**Selected Groups Present**	Wednesday, Tue tbc
Week 6	*Data Analysis*	Cluster Analysis ii	**Selected Groups Present**	Wednesday, Tue tbc
Week 7	Reading Week			No lectures, workshops
Week 8	*Data Analysis*	Association Rule Mining	**Selected Groups Present**	Wednesday, Tue tbc
Week 9	*Data Analysis*	Data Warehousing	**Selected Groups Present**	Wednesday, Tue tbc
Week 10	*Data Analysis*	Data Warehousing	**Selected Groups Present**	Wednesday only
Week 11		Cube Computation		Wednesday only
Week 12		Revision & Exam Prep		